NSM NORDIC STATISTICAL MEETING 2026

Deep learning for transparent production of official statistics

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Content

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  deep learning with reporting uncertainty

- Applications with examples:
  - synthetic datasets for confidentiality protection
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- Conclusions

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Motivation

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- Requirement of official statistics:

  interpretability and transparency of *results* and *methods*

- Critical domains which need:
  - efficient and up-to-date *solutions*
  - informative and honest *reporting*

- Advanced modeling and inference *tools* (e.g. deep-NN, Bayesian methods): computational feasible!

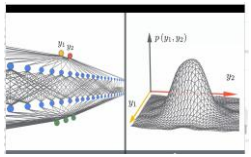
Measuring uncertainty of Deep Learners

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➢ **Analytic tools** to **understand/evaluate** DL, e.g. ***GP ⇔ BNN*** :

[*Deep **N**eural **N**etworks as **G**aussian **P**rocesses,* Lee, J. etal (2018) *https://arxiv.org/abs/1711.00165*]

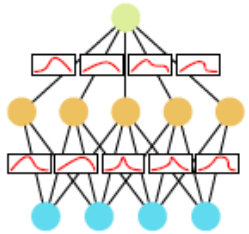
https://en.wikipedia.org/wiki/Neural_network_Gaussian_process#cite_note-:1-4



➢ **Variation techniques** for *DL/LLMs*

*[Look Before You Leap: An Exploratory Study of **Uncertainty Measurement** for Large Language Models,*

Huang, Y. etal, (2024) https://arxiv.org/abs/2307.10236]



from: *https://arxiv.org/abs/2007.06823*

Q =>

- M1: **single** inference (max/avg prob. or max/avg entropy) → Answer A1
- M2: **Stochastic** inference/sample based/**model** variation (e.g. dropout, deep ensembles) (VR, VR0) → Answers A2.1, A2.2, …
- M3: **Stochastic**/**Data** Perturbation (maxDiff VR, MaxDiff VR0) → Answers A3.1, A3.2, …

➢ **Bayesian methods** ← **weights' distributions** (positive example: *autoBNN in Tensorflow Probability, using **GP ⇔ BNN***)

- mitigate risks arising from overly confident yet incorrect predictions made by LLMs
- provide uncertainties over predictions, which can enrich decision-making
- enable the use of domain-knowledge priors

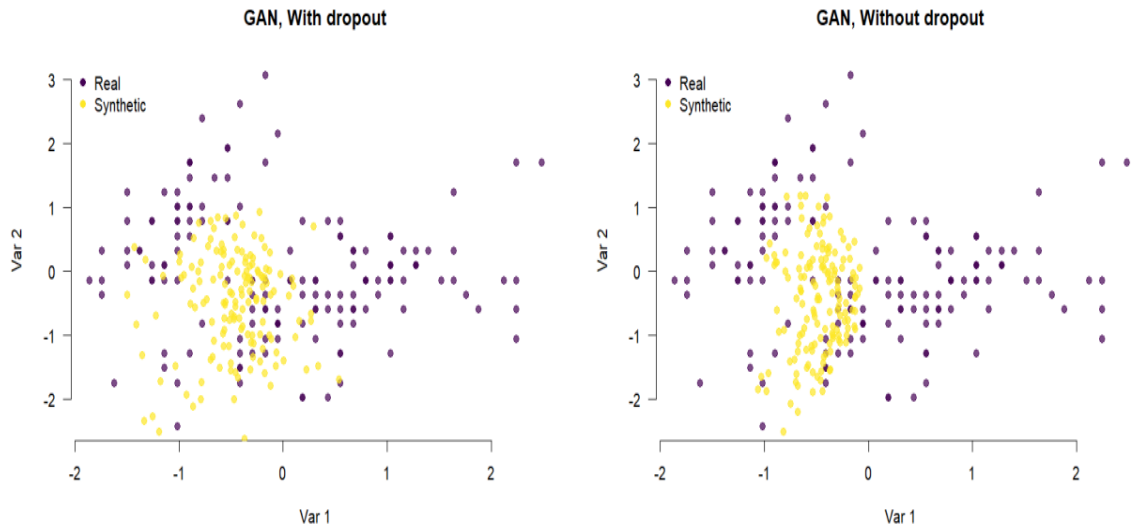


[(2024) *Position: **Bayesian Deep Learning** is Needed in the Age of Large-Scale AI,* Papamarkou, Th. Etal, https://arxiv.org/abs/2402.00809]

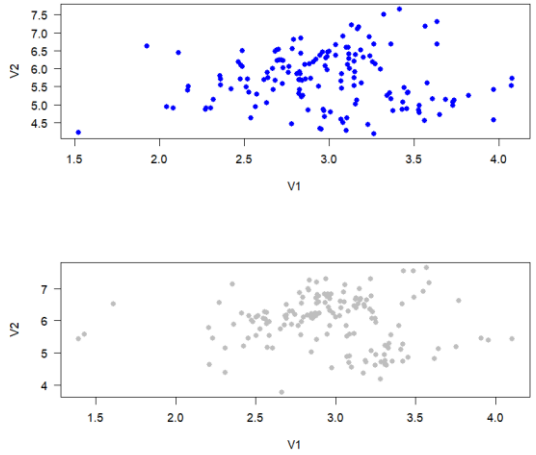
Applications: synthetic data for SDC

- Generative Adversarial Networks (GAN) with MC dropout regularization (⇔ *Bayesian NN*) and Bayesian nonparametric mod.

(with *RGAN*)



Bayesian synthetics
(with *synMicrodata*)

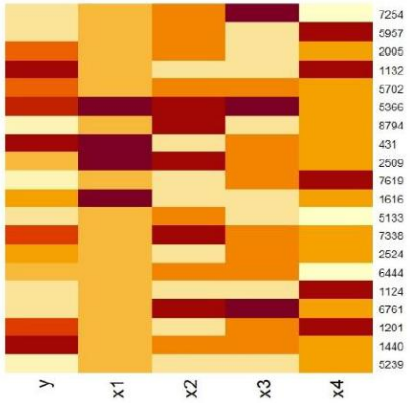


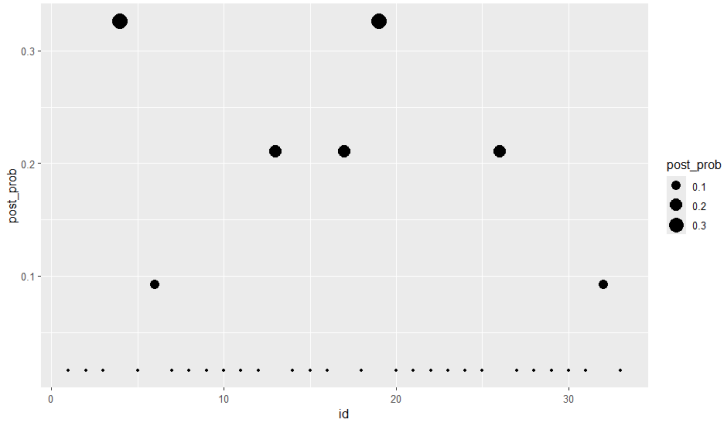
Applications: automatic anomaly detection

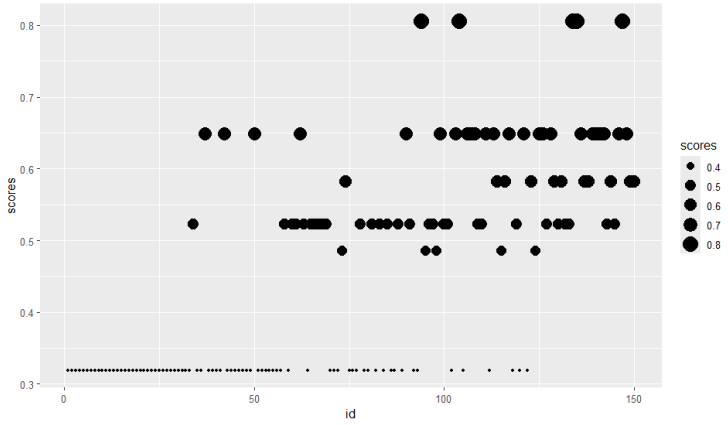
(for: validation and SDC unsupervised testing)

Rare patterns
Anomaly scores and
Anomaly *posterior probabilities*

(with *d.i.f.*, *HRTnomaly, h2o*)

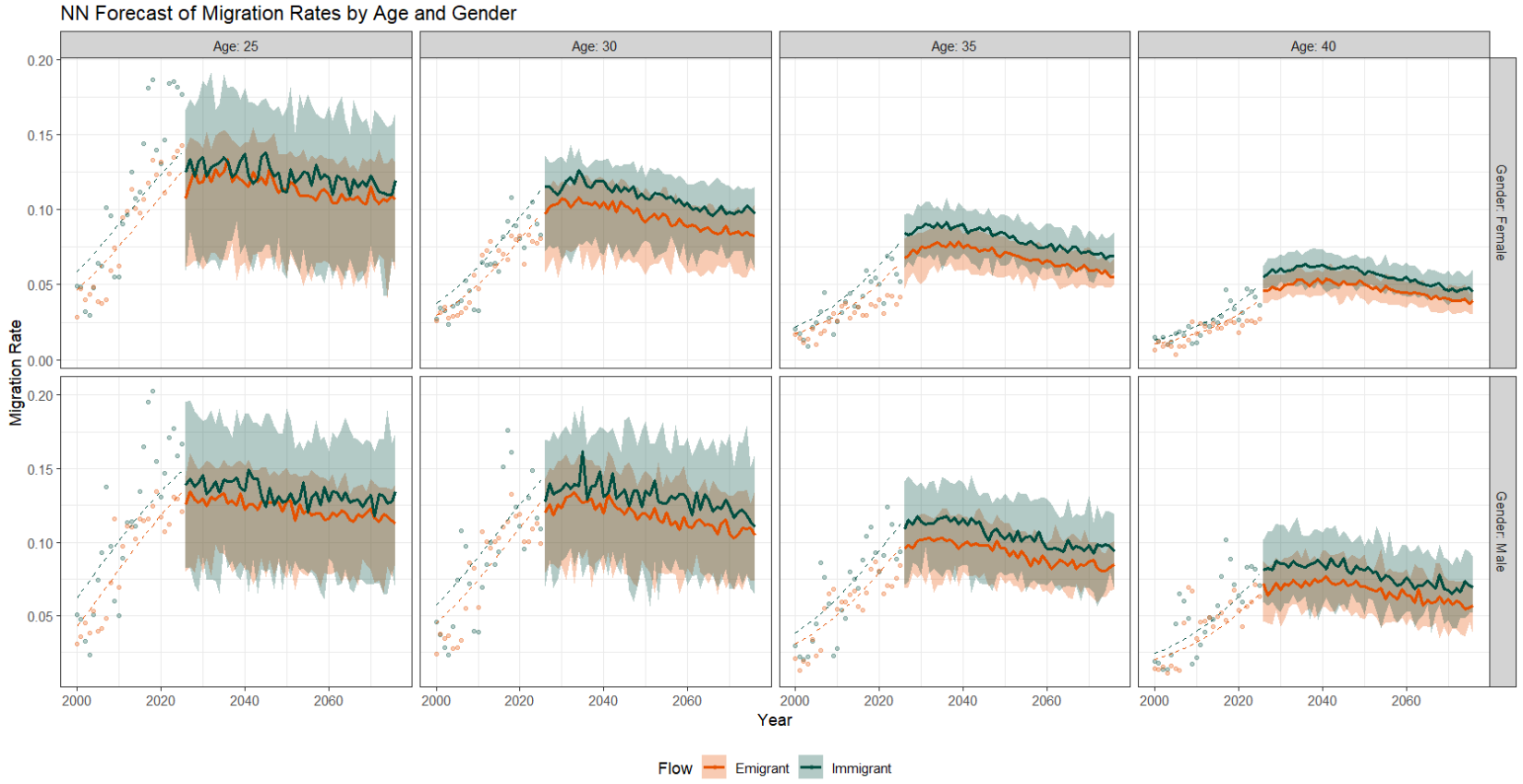


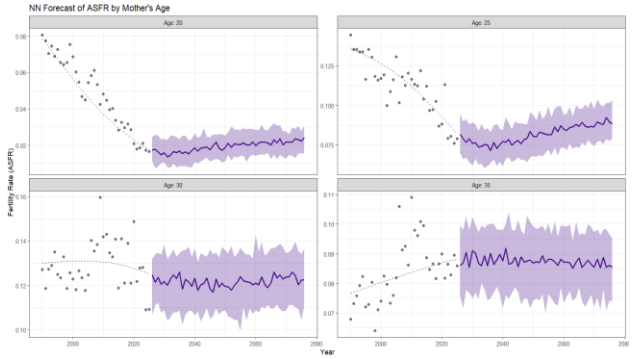


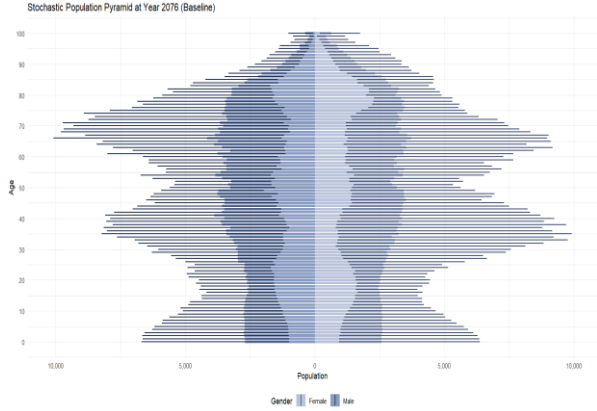


Applications: time series forecasting with (B-reg) NN

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Conclusions

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Reporting uncertainty of the statistical results produced by deep (NN) learners increases reliability, transparency and trust.

Useful links

- *SIPP (Time Series/Forecasting for population projections). GitHub Repository. URL: https://github.com/violetacln/SIPP*
- *Testing SDC Tools. GitHub Repository. URL: https://github.com/violetacln/testingSDCtools*
- *SDMX_ML_Validation. GitHub Repository. URL:* https://github.com/violetacln/sdmx_ML_validation

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